AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to computer graphics

Final Examination
Spring Semester 2024/2025

Date: 2/June/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: This part consists of 10 multiple choice questions. For each question, encircle the label a, b, c, or d as correct answer. Each question carries 2 marks.

1 The image class ..... is used to represent an image with an immediately available data store.

A LookupOp
B LookupOp
C BufferedImage
D Graphics2D

2 ..... produces series of rendered images (frames) that depicts the change in a scene

A None of the above
B None of the above
C Animation
D None of the above

3 The number of parameters used to construct a Timer object is .....

A 3
B 2
C 1
D 4

4 .....is known as local or modeling coordinate system.

A World space
B View space
C Device space
D Object spac

5 .....of the curve states that it should not only be continuous without jumps, it should have no sharp bends.

A Controllability
B Locality principle
C Smoothness
D None of the above

6 The center of the rectangle defined by RoundRectangle2D.Double(a1,a2,a3,a4,a5,a6) is at .....

A (a1+0.5a5,a2+0.5a6)
B (a3,a4)
C (a1+0.5a3,a2+0.5a4)
D (a5,a6)

7 At time 15:15:3 the clock arm representing the hour should make an angle .....with the positive vertical axis.

A 90
B 97.5
C 5
D 15

8 the function .....is a member function of the GeneralPath class and is used to check whether a given point in inside a given GeneralPath.

A xor
B contains
C intersect
D None of the above

9 .....is used to find the common points between two areas in java.

A exclusiveOr
B Subtract
C Add
D Intersect

10 The color space that is common on monitors is .....

A RGB
B CYMK
C CIEXYZ
D None of the above

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Part II: This part consists of four questions with a total of 80 Marks, solve all questions.

Question 2 [20 Marks]

a. Explain the concept of composite transforms and how they can be used to create complex transformations in Java 2D.

b. Write a java code segment to draw the following Latin letter:



You must use suitable Stroke and GeneralPath.
Flip the symbol vertically as in the figure:



Question 3 [20 Marks]

a. Explain the purpose and method of creating an Area object in Java.

b.

Having the following code along with its output (adopting default font size and face):
g.drawString("Sample output from TM240 code.", 100, 100);

Modify the above code to output this shape:



Question 4 [20 Marks]

a. Discuss the significance of the controllability principle in curve design. How does it guide users in modifying curves to achieve specific shapes?

b. Having an image object of type BufferedImage that has the following artifacts:

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1) image is sheared towards the negative x-directions
2) image is sheared towards the negative y-directions
3) image is rotated clockwise around its center
4) image is zoomed in
5) image is to be written to the hard disk

Write a java code segment to resolve the above problems.

Question 5 [20 marks]

a. Explain the limitations of Bézier curves for higher-degree polynomials and why they are not suitable for large sets of control points.

b. Write a function in java code to implement Mandelbrot fractal sequence with maximum number of iterations of 64 for an arbitrary setting of the imaginary constant

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End of questions

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